

Role: Exam Security Engineer

Client : CUPA

Location : Cambridge (3 days a week)

Type : Contract

Job Summary

We are seeking a skilled Exam Security Engineer to enhance the Safe Exam Browser (SEB) used for secure online assessments. The role focuses on advancing SEB's technical configurations, developing detection and authentication features, and implementing robust procedural and auditing measures to support exam integrity. This position requires a blend of software development expertise, security analysis, and collaboration with exam centre operations to maintain the security of high-stakes testing environments. The ideal candidate will drive ongoing maintenance, iterative updates, and auditing processes to ensure SEB remains robust against emerging threats, including advanced authentication mechanisms, logging systems, cryptographic protections, process security controls, virtualization safeguards, and hardware monitoring capabilities.

Required Skills and Qualifications

- **Technical Skills:**
 - Proficiency in frontend software development, particularly in languages and frameworks used for browser-based applications (e.g., JavaScript, HTML5, CSS3, React, or similar for browser extensions and user interfaces).
 - Strong knowledge of browser security concepts, including client-side scripting, DOM manipulation, cryptographic APIs (e.g., Web Crypto API for headers and signatures), and browser extension development.
 - Experience with browser security tools, web-based detection techniques, logging frameworks in JavaScript, and anti-cheating features in web applications for educational or high-stakes environments.
 - Familiarity with web technologies and tools for client-side interactions (e.g., Web APIs, Service Workers), as well as secure session management and auditing in browsers.
- **Security and Analytical Skills:**

- Expertise in client-side vulnerability assessment, threat modelling, and mitigation strategies for web-based proctoring software, including iterative development for cryptographic and logging features in the frontend.
- Ability to design and implement frontend technical controls to address potential security vectors (e.g., external scripts, session integrity).
- Strong problem-solving skills with a focus on proactive maintenance, updates, and intelligence-building through client-side data analysis.
- **Soft Skills:**
- Excellent communication skills for collaborating with non-technical stakeholders (e.g., exam invigilators, centre staff), as well as documenting procedures.
- Ability to prioritise tasks, such as fast-tracking frontend feature development like cryptographic interfaces.

Experience

- Minimum 3-5 years in frontend software development, preferably in edtech, proctoring, or secure web application development.
- Hands-on experience with browser lockdown tools or extensions (e.g., SEB, Respondus LockDown Browser), including config management and client-side logging integrations.
- Proven track record of developing and maintaining frontend security features that require ongoing updates, such as web-based cryptographic systems and auditing interfaces.
- Experience in exam centre operations, invigilation protocols, or collaborative development with vendors is a plus.